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GOOD PRACTICE IN AI FOR EDUCATION: Spotlight on EU Case Studies and Insights

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WHAT IS THE MAPPING'S CONTEXT AND PURPOSE

The European Commission's Directorate-General for Education, Youth, Sport and Culture (DG EAC) commissioned this mapping to address the rapid rise and growing influence of Artificial Intelligence in European educational systems. Whilst AI technologies are increasingly present in classrooms, ranging from adaptive learning platforms to automated assessment tools, considerable disparities persist in access, adoption, and pedagogical integration across Member States. Many educators report feeling unprepared and insufficiently supported or trained to use AI effectively, highlighting the need for coherent policy guidance and practical support at all levels of education.

This mapping focuses on the identification of good examples of the ethical and responsible use of AI in education across the EU at all levels of education. It highlights the most promising and innovative examples whilst encouraging and supporting ethical AI adoption among teachers, learners, and institutions.

HOW DID WE SELECT THE GOOD PRACTICES?

The identification of good practices followed a mixed-method approach combining extensive desk research, survey data, and qualitative interviews. A multilingual desk review screened academic publications, project databases, and grey literature, leading to the identification of 110 verified practices. A pan-European online survey targeting educators, school leaders, policymakers, and other stakeholders gathered 1,306 responses, from which 99 additional good practice examples were extracted.

GOOD PRACTICE IN AI FOR EDUCATION: Spotlight on EU Case Studies and Insights

From the combined set of 209 practices, fourteen cases were selected for in-depth investigation based on their pedagogical core, human-centred dimension, and implementation reality. Semi-structured interviews were conducted with those involved in development and implementation to explore key enablers and barriers, as well as impact. The mapping was conducted from January to October 2025.

KEY SURVEY FINDINGS

The survey revealed that generative AI systems are the most widely used tools, followed by adaptive and intelligent tutoring systems and AI-based assessment and feedback platforms. More than half of respondents reported increased teaching efficiency, with additional benefits in pedagogical innovation and student engagement. However, the findings also underscore significant challenges: teacher training and support emerged as the most critical success factor for the successful application of AI, yet many teachers reported either a complete absence of policy support or uncertainty about whether such support existed. This disconnect between policy development and classroom practice highlights the urgent need for systematic evidence on both what works in AI implementation and the support mechanisms required to enable it.

HOW TO READ THE PRACTICE SUMMARIES

Each good practice is presented using four dimensions that help explain why it works well:

- "The practice at a glance" section provides an overview of what the practice is, where it operates, and what it aims to achieve.
- "How it improves teaching and learning" section explains why this practice benefits teaching and learning. This covers how it supports educational goals, what impact it has on students, and how it helps achieve better learning outcomes.
- "How It Supports Teachers and Learners" section explains why this practice works well for the users involved. This includes how it affects teacher workload, how teachers and students were involved in its design, and how it supports all learners including those with additional needs.
- "Making it work in practice" section explains what makes this practice work in real-world conditions. This covers practical aspects such as data protection, costs, technical requirements, and whether it can be sustained or used elsewhere.

The following pages present summaries of each good practice identified through this mapping exercise.

Language and literacy platform, Intelligent and adaptive tutoring systems (ITS), Learning analytics and progress monitoring

Lalilo

France | *Early Childhood and Primary Education*

“Teachers reported that pupils often ask to use the platform, seeing it as enjoyable rather than remedial work. This helps struggling readers stay engaged long enough to make real progress.”

KEY BENEFITS

Personalised Learning

Teacher Time-Saver

Confidence Building

THE PRACTICE AT A GLANCE

Lalilo is an adaptive, web-based reading platform used across the first cycle of French primary education. The platform offers personalised phonics, word recognition, and comprehension activities adapted to each pupil's level. Its main goal is to build foundational reading skills for learners from kindergarten through the second year of primary school. Teacher engagement with the platform has been encouraged through the P2IA programme (Partenariats d'innovation et intelligence artificielle), a French Ministry of Education initiative fostering collaboration between EdTech companies, researchers, and educators to ensure tools are sound and fit for the classroom.

A 2023 randomised controlled study of first-grade students found a statistically significant impact on oral reading fluency, with the strongest positive effects for the students who needed most support.

HOW IT IMPROVES TEACHING AND LEARNING

Lalilo addresses a common challenge in priority education classes: pupils arrive with very different reading levels, and some still struggle with basic letter recognition in later grades. The platform quickly identifies where each child is and proposes activities to address their specific gaps. Content follows research-based reading progression and aligns with both national and European basic skills priorities.

A 2023 impact study found clear improvements in oral reading fluency, with the strongest positive effects for students who needed the most support. Teachers reported that pupils often ask to use the platform, seeing it as enjoyable rather than remedial work. This helps struggling readers stay engaged long enough to make real progress. The platform's design lets students work independently, and errors are treated as a normal part of learning rather than failures, helping to rebuild confidence for those who find reading difficult.

Language and literacy platform, Intelligent and adaptive tutoring systems (ITS), Learning analytics and progress monitoring

Lalilo

France

ECEC-Primary

HOW IT SUPPORTS TEACHERS AND LEARNERS

Lalilo follows Universal Design for Learning principles. Speech recognition lets students interact using their voice, and extra vocabulary support is available for pupils who do not speak French at home. These features help make the platform accessible to a wider range of learners.

The platform reduces teacher workload by tracking each student's progress through dashboards. Teachers describe it as a time-saver: marks and progress reports are generated automatically, freeing them to focus on targeted support rather than manual checking. The data also makes handovers between teachers easier, as detailed information on each child's achievements can be shared during transitions.

Teachers have been involved in Lalilo's development through an ambassador programme where they provide feedback and help create content. Schools hold weekly meetings to address parental questions about digital tools, though the platform remains underused at home, possibly due to limited device access or concerns about screen time.

MAKING IT WORK IN PRACTICE

Lalilo's use through the P2IA programme required integration with the GAR (Resource Access Manager), which ensures pupil data stays under the control of the education authority rather than the company. This provides strong data protection, though both teachers and school leaders noted that the system can cause technical slowdowns that sometimes disrupt lessons.

The school studied benefited from a supportive environment: funding for tablets through the now-discontinued "École Connectée" programme, a responsive technical support team from Lalilo, and guidance from a Digital Education Advisor. These elements were essential for making the platform work well.

The main challenge for wider use has been dependence on specific funding and good local infrastructure. However, renewed support and targeted investment are now helping more schools adopt similar practices across France.

Website: https://www.lalilo.com/

TEACHER CHALLENGES & INSIGHTS

Technical Slowdowns

Funding Dependence

Limited Home Use

> *“The platform reduces teacher workload by tracking each student’s progress through dashboards. Teachers describe it as a time-saver: marks and progress reports are generated automatically, freeing them to focus on targeted support rather than manual checking.”*

DigiHavel

Czech Republic | *Secondary Education*

> *"A key strength is that DigiHavel teaches students about AI itself, not just civic topics. Students learn that AI can produce inaccurate information [...] so they must think critically, check facts, and verify answers as part of their learning."*

THE PRACTICE AT A GLANCE

DigiHavel is a free digital tool developed by Responsible Citizenship, an NGO in the Czech Republic, to support civic education using AI. Designed for teachers at the secondary stage of primary school, it introduces students to the principles of democracy, contrasts it with totalitarianism, and encourages reflection on fundamental human rights. Through collaborative activities, learners tackle contemporary social issues whilst engaging with artificial intelligence. The platform offers five ready-to-use civic education lesson plans developed by experts from Masaryk University's Faculty of Education, aligned with the Competence Model for Responsible Citizenship. A built-in chatbot inspired by the first Czech president Václav Havel supports classroom activities and helps students explore democratic values.

The chatbot's knowledge base draws on verified data from the Library of Václav Havel, validated by people who knew him personally. Havel's family gave their approval before development began.

KEY BENEFITS

- **Critical AI Literacy**
- **Ready-to-Use Resources**
- **Inclusive Design**

HOW IT IMPROVES TEACHING AND LEARNING

DigiHavel uses a four-step competence model: knowledge acquisition, working with and verifying information, forming personal opinions, and taking action. The tool attracts students who are already comfortable with digital devices and AI, and teachers report that students actively participate because of the fresh approach to learning.

A key strength is that DigiHavel teaches students about AI itself, not just civic topics. Students learn that AI can produce inaccurate information through "hallucination", so they must think critically, check facts, and verify answers as part of their learning. Teachers find the tool works best for students aged thirteen and fourteen, particularly in teaching topics such as democracy, totalitarianism, human rights, and AI. The active, inquiry-based approach is valuable in an education system where lecture-based teaching still dominates, helping move classrooms towards more engaging methods.

DigiHavel

Czech Republic | *Secondary Education*

TEACHER CHALLENGES & INSIGHTS

- **Technology Adoption for Older Teachers**
- **Unreliable Internet Access**
- **Unstable Funding**

> *"The active, inquiry-based approach is valuable in an education system where lecture-based teaching still dominates, helping move classrooms towards more engaging methods."*

HOW IT SUPPORTS TEACHERS AND LEARNERS

DigiHavel significantly reduces, once and if familiar with the tool, the teacher workload by providing a complete package of resources. Over three hundred schools across the Czech Republic now use the platform thanks to these teacher-friendly materials. Once familiar with the tool, teachers find preparation time drops considerably because the lesson plans are well designed and easy to follow. The chatbot sometimes provides feedback on student work, supporting rather than replacing teacher input.

The tool enables varied teaching methods including small group work, individual activities, and whole-class discussions, moving beyond traditional lectures. It also supports inclusive classrooms: non-Czech speaking students, including Ukrainian refugees, can switch the tool to English, and a text-to-speech feature helps students with dyslexia or dysgraphia who struggle with long written texts. However, not every student has reliable internet access, which can make individual use difficult in some cases. Older teachers have sometimes found the technology challenging to adopt.

MAKING IT WORK IN PRACTICE

The development team worked closely with experts and stakeholders, running three phases of piloting and testing with different groups of teachers to ensure the tool works well in classrooms. The team consulted Václav Havel's family for approval and obtained reliable data from the Library of Václav Havel, validated by people who knew him personally. Ethical considerations were central from the start, given that the tool draws on a real person. To make clear that DigiHavel is AI inspired by Havel rather than an accurate representation, the graphical design uses a caricature rather than a realistic image, encouraging critical thinking about what is real and what is not.

The tool has scaled successfully from initial pilots to several hundred schools, and the team is already developing new AI assistants using the same approach. However, sustainability depends on ongoing funding. Keeping the tool relevant requires constant updates, and as more teachers use it, operating costs rise. Currently, the team relies heavily on pro bono work and below-market rates whilst keeping the tool free for educators. Securing stable funding remains a key challenge.

Website: https://odpovedneobcanstvi.cz/digihavel/

NOLAI

Exploratory learning environments; Language and literacy platforms; Formative assessment and real-time feedback; Learning analytics and progress monitoring; Immersive content creation (AR/VR)

The Netherlands | *Primary Education*

"Unlike initiatives that start with technology, NOLAI projects begin with concrete classroom challenges identified by teachers themselves. This ensures the solutions are genuinely useful in day-to-day teaching."

KEY BENEFITS

Teacher-Led Innovation

Equity Focus

Open Source Commitment

THE PRACTICE AT A GLANCE

NOLAI is a co-creation laboratory hosted by Radboud University that develops and tests classroom-ready AI solutions together with schools, researchers, and companies. The lab works through time-limited projects with clear research questions, classroom pilots, and plans for scale-up. Two primary education pilots illustrate this approach: one uses short, teacher-guided virtual reality sessions to immerse pupils in rich scenes that support vocabulary learning; the other embeds sensors in Montessori arithmetic materials to capture how pupils interact with tasks, with an AI tool providing teachers with insights into each learner's process.

Of the seventeen projects selected so far, twelve come from secondary education, five from primary, and in 2025 two requests came from special education. The standard budget is €300,000 per project over three years, funded through the National Growth Fund.

HOW IT IMPROVES TEACHING AND LEARNING

NOLAI's strength lies in its problem-first approach. Unlike initiatives that start with technology, NOLAI projects begin with concrete classroom challenges identified by teachers themselves. This ensures the solutions are genuinely useful in day-to-day teaching.

The VR vocabulary project addresses a specific equity issue: children who struggle with language because they lack real-world experiences. Recognising that context is essential for learning, the team chose VR to provide an immersive alternative, letting students virtually visit a beach or see the true scale of a giraffe. The Montessori sensors project responds to teachers' difficulty monitoring individual learning in busy classrooms. By embedding sensors in arithmetic materials, the AI tool gives teachers real-time information about how each student is working and highlights key moments in their learning, helping teachers intervene with the right support at the right time. Early pilots show strong student engagement, though full impact data is still being collected.

NOLAI

Exploratory learning environments; Language and literacy platforms; Formative assessment and real-time feedback; Learning analytics and progress monitoring; Immersive content creation (AR/VR)

The Netherlands

Primary Education

TEACHER CHALLENGES & INSIGHTS

Initial Teacher Hesitancy

Teacher Shortages and Mobility

Mismatched Timelines

"By involving educators as co-designers rather than simply end-users, the lab builds trust and ownership. [...] Structured support and practical materials, not just funding, proved to be the main drivers of sustained teacher engagement."

HOW IT SUPPORTS TEACHERS AND LEARNERS

NOLAI is built around teacher-led problem definition. Schools across the country identify challenges, and "teachers in residence", selected through open calls, provide feedback on projects based on their classroom experience. All projects are developed jointly by schools, researchers, and companies, with teachers involved in both design and implementation across the typical three-year project duration.

This collaborative model helps address teacher reluctance about AI. By involving educators as co-designers rather than simply end-users, the lab builds trust and ownership. Targeted training on basic AI concepts and practical classroom use supports this, proving especially important for primary school teachers who were initially hesitant about using AI tools.

MAKING IT WORK IN PRACTICE

Five co-creation managers with varied backgrounds gather questions from schools across the country, including primary, secondary, and special education. They connect these questions with relevant researchers and businesses, coordinating project proposals through structured sessions. Each project establishes a data management plan at the outset, with research partners typically handling data rather than schools. An ethics team works closely with developers, teachers, and scientists to ensure responsible AI use.

Scalability is a selection criterion for projects. All developments must be open source and usable by other companies, reducing dependence on single providers. The standard budget is €300,000 per project over three years, funded through the National Growth Fund. Challenges include teacher shortages and mobility, as well as different working timelines across businesses, researchers, and schools. Structured support and practical materials, not just funding, proved to be the main drivers of sustained teacher engagement.

Website: https://www.ru.nl/en/nolai

Piloting AI-Powered Virtual Assistance in Schools

Intelligent and adaptive tutoring systems (ITS); Learning analytics and progress monitoring

Italy

Vocational Education and Training

> *"Comparing results between classes using AI systematically and those using innovative methods without AI showed that the AI-supported classes improved more. All students in the AI group progressed to the next year, whilst some in the comparison group did not."*

KEY BENEFITS

- **Measurable Impact**
- **Integrated Approach**
- **Comprehens ive Training**

THE PRACTICE AT A GLANCE

This national pilot launched by the Italian Ministry of Education and Merit brings AI-powered virtual assistance into fifteen secondary school classes across four regions: Lombardy, Abruzzo, Marche, and Tuscany. The pilot runs in both lower and upper secondary schools, including vocational institutions. It explores how AI can enhance teaching and learning, with an initial focus on STEM subjects and Italian language. The initiative also aims to reduce educational gaps by providing targeted support to students who need it most.

The school uses three distinct AI tools, each serving a different purpose: Esercizi Guidati for personalised feedback, Gemini for teacher support, and Notebook LLM for advanced content generation. Free licences were provided by Google under an agreement with the Ministry of Education.

HOW IT IMPROVES TEACHING AND LEARNING

The pilot aims to personalise learning to help students achieve more, particularly within the shorter four-year vocational curriculum. The idea is that AI can help create individual learning paths for each student, acting like an additional teacher by providing personalised feedback and giving educators better insight into each student's progress.

Google Classroom serves as the main platform for delivering content and communicating with students and parents. Four AI tools serve different purposes: Esercizi Guidati (Guided Exercises) for personalised feedback during independent practice; YouTube videos with built-in self-assessment; Gemini for teacher support and content creation; and Notebook LLM for more advanced content generation. Comparing results between classes using AI systematically and those using innovative methods without AI showed that the AI-supported classes improved more. All students in the AI group progressed to the next year, whilst some in the comparison group did not.

Intelligent and adaptive tutoring systems (ITS); Learning analytics and progress monitoring

Piloting AI-Powered Virtual Assistance in Schools

Italy

Vocational Education and Training

TEACHER CHALLENGES & INSIGHTS

Initial Adjustment Period

Data Protection Complexity

Single Provider Dependence

“As the principal put it, the focus is not on whether students use AI, but on using it ethically, critically, and effectively.”

WHY DOES IT WORK WELL FOR TEACHERS AND LEARNERS?

The school took a thorough approach to training. Teachers received specific preparation from Google; in addition, as a national training hub through the National Recovery and Resilience Plan, the school was able to offer professional development through online courses for beginners and advanced users, alongside in-person sessions on AI in teaching. All new staff complete a mandatory twenty-five-hour induction covering Google Workspace tools and innovative teaching methods.

Teachers built AI tools directly into daily practice, so students became familiar with them through regular use across subjects. Feedback has been largely positive after an initial adjustment period. Teacher cooperation, student enthusiasm, and sharing ideas across classrooms encouraged experimentation. An internal survey examining how students already use AI on their own reinforced the need to guide them toward informed and critical use. As the principal put it, the focus is not on whether students use AI, but on using it ethically, critically, and effectively.

MAKING IT WORK IN PRACTICE

Training began in September 2024, with students starting to use the tools by November of that year. Google provided free licenses under an agreement with the Ministry of Education. Data privacy was addressed carefully from the start: the initiative began with teachers only, to minimise risk, and all participating teachers gave explicit consent. Student data protection followed privacy laws fully. All data stayed within the EU, and the enterprise Google licenses ensured data is not reused for AI training.

The five Tuscan pilot schools stay in regular contact through online meetings, shared training sessions, and a WhatsApp group including school leaders and regional education officials. This allows them to coordinate efforts, jointly deciding what data to track and symbolically launching the pilot on the same day. The school is evaluating impact by comparing results between experimental and control classes. Future goals include helping students use AI tools independently for critical thinking across subjects, which will require careful work with data protection officers as the project expands.

Dialogue-based tutors; Exploratory learning environments

CurreChat

Finland

Higher Education

“Students found these simulations so lifelike that they felt genuine urgency to find the right answers. The ability to practice in a safe space where mistakes become learning opportunities proved especially valuable for high-achieving students.”

KEY BENEFITS

Safe Practice Environment

Creative Exploration

Institutional Security

THE PRACTICE AT A GLANCE

CurreChat is a generative AI tool developed specifically for the University of Helsinki community, giving students and staff secure access to large language model capabilities. Based on OpenAI models hosted within Microsoft Azure's cloud service in the EU, the platform ensures data protection whilst being accessible through a university portal. Its integration with university systems allows teachers to customise CurreChat for specific courses, a feature particularly valuable for teaching. The University of Helsinki has taken an encouraging stance on generative AI, viewing AI literacy as an important skill for working life.

Since its launch in autumn 2023, over 260 courses at the University of Helsinki are using CurreChat. In medical education, patient characters were designed to sometimes provide less-than-perfect or even misleading information, deliberately mimicking real-world consultations.

HOW IT IMPROVES TEACHING AND LEARNING

CurreChat directly supports the university's policy on AI in teaching, providing a secure and customisable platform for course use. The tool has enabled novel teaching approaches, including two AI-powered characters: Tero, a realistic forest owner who serves as an information source during role-play exercises, and Madida, a supportive learning companion guiding students through planning exercises.

In medical education, a faculty member developed patient characters allowing future doctors to practise online consultations, asking the right questions and experiencing realistic interactions. Students found these simulations so lifelike that they felt genuine urgency to find the right answers. The ability to practise in a safe space where mistakes become learning opportunities proved especially valuable for high-achieving medical students who tend to be perfectionists. Students report that CurreChat sparks creativity, offers ideas they would not have thought of alone, and allows them to explore topics through questions.

Dialogue-based tutors; Exploratory learning environments

CurreChat

Finland

Higher Education

HOW IT SUPPORTS TEACHERS AND LEARNERS

Development involved close work with teachers and subject experts. A survey of sixty-six educators first identified a range of needs for AI in education, including AI characters, student learning support, and teacher assistance. Through testing with experts, both characters addressed nine of twelve identified teacher needs, showing strong potential for improving higher education. Development combined insights from AI specialists and learning scientists, whilst ensuring data handling follows GDPR without user information being passed on or used for model training.

Strong support from university leadership, including the Vice Rector and Faculty Dean, gave staff freedom to innovate. CurreChat is now free and accessible to the University of Helsinki community, promoting fair access to AI-supported learning. Student feedback has been largely positive, with one course-level faculty survey showing ninety percent of students recommending continued use. Both students and staff acknowledge initial nervousness about new tools, with some finding it easier not to engage, yet there is general openness to trying new approaches.

MAKING IT WORK IN PRACTICE

Development included work with university legal experts to ensure data security and user privacy. Technical issues can be shared with IT support for development and troubleshooting purposes, while user prompts and identifying data remain protected. Challenges include managing technical issues when systems crash during class and ensuring the platform can scale as it becomes more popular. A support team has handled over 350 queries about CurreChat, though some staff report unclear support structures when problems arise.

The platform has very limited use beyond the university, permitted only in specific educational cases. Sustainability depends on continued institutional support and clear maintenance arrangements. Staff note that whilst the tool improves teaching, it requires significant preparation time for developing characters and scenarios, a common challenge for innovative AI teaching that needs dedicated investment before students can benefit from seemingly instant feedback.

TEACHER CHALLENGES & INSIGHTS

- **Significant Preparation Time**
- **Technical Crashes During Class**
- **Initial Nervousness with New Tools**

> *"The tool requires significant preparation time for developing characters and scenarios, a common challenge for innovative AI teaching."*

Generative AI for content creation; Assistive technology for special educational needs; Language and literacy platforms

@2SOSUAI

Denmark

Vocational Education and Training, Adult Learning

"Rather than giving direct answers, Amina acts as a round-the-clock learning partner, engaging students in reflective conversation and encouraging critical thinking."

KEY BENEFITS

- **Targeted Inclusion**
- **Reflective Learning**
- **Accessible Design**

THE PRACTICE AT A GLANCE

This project explores how generative AI can help female students with Danish as a second language succeed in social and healthcare education. It specifically supports women from diverse cultural and linguistic backgrounds who often struggle to complete their training due to language barriers and unfamiliarity with Danish cultural norms. Twelve social and healthcare schools across Denmark are involved. The project is based on the idea that generative AI can help address the challenges bilingual women face in language learning, professional skill development, and active participation. The educational chatbot Amina is the main focus of our analysis.

HOW IT IMPROVES TEACHING AND LEARNING

Amina was developed to support women facing limited Danish skills and unfamiliarity with everyday cultural practices. Drawing on input from students and teachers, it was introduced before student placements to provide relevant guidance. The chatbot is trained according to government educational goals for social and healthcare assistants, including placement requirements.

Rather than giving direct answers, Amina acts as a round-the-clock learning partner, engaging students in reflective conversation and encouraging critical thinking. Built-in buttons offer options like understanding assignments without direct answers, reflecting on placement scenarios, and visualising real-life situations. A feature explaining Danish words with examples helps both non-native speakers and those unfamiliar with technical terms.

Students show clear progress over time. Initially they asked simple questions looking for quick answers, but gradually they engage more critically, rephrasing questions for clearer responses and requesting realistic scenarios to practise skills. Some use Amina for self-assessment during placements, particularly for communication with elderly residents, exploring familiar meal preparation, and learning about dementia care.

Generative AI for content creation; Assistive technology for special educational needs; Language and literacy platforms

@2SOSUAI

Denmark

Vocational Education and Training, Adult Learning

TEACHER CHALLENGES & INSIGHTS

Language Barriers in Introduction

Sensitive Data Incident

Pedagogical Shift for Teachers

“Students show clear progress over time. Initially they asked simple questions looking for quick answers, but gradually they engage more critically, rephrasing questions for clearer responses and requesting realistic scenarios to practise skills.”

HOW IT SUPPORTS TEACHERS AND LEARNERS

At each participating school, two teachers are selected to work with IT consultants on development. The AI tools were created through Learning Factory workshops bringing students and teachers together, guided by researchers. The Welfare Technology Agency showed applicable AI technologies, and schools with similar ideas collaborated.

The IT consultant introduces Amina in class with teachers present, starting with a general AI introduction and practical examples, encouraging students to approach the chatbot with curiosity and reflection. Language barriers are a significant challenge since some students are not fluent in Danish, requiring highly visual presentations with hands-on exploration time. Teachers receive training on using Amina and are designing tasks that include the chatbot, moving from traditional teaching to learning experiences with AI as a partner.

MAKING IT WORK IN PRACTICE

The project runs across twelve schools in partnership with Aalborg University's Centre for Educational Research, the Centre for Application of IT in Vocational Education and Training, and the Knowledge Centre for Welfare Technology East and West Denmark. An advisory board of eight members ensures the project stays relevant and practical. Funding comes from the VELUX Foundation.

Amina runs on ChatGPT with tailored prompts, with subscription costs covered by the Centre for IT in Education. As an open-source tool, students access Amina through a simple link without needing to log in, ensuring a secure environment that follows data protection rules. One incident involving sensitive medical data led to improvements in the chatbot's introductory messages. The IT consultant provides Amina with a weekly update based on feedback, tailoring it through an overview of anonymised student interactions. The Welfare and Technology Department allocated fifty to seventy-five hours for continued development and plans to expand Amina across all twelve Danish social and healthcare schools given its success.

Generative AI for content creation; AI-powered curation and recommendation

KI-Pilotschulen (AI Pilot in Schools)

Austria

Primary and Secondary Education

"Research shows that teacher confidence or anxiety significantly affects how AI is adopted, which justified this teacher-focused approach."

KEY BENEFITS

Strategic Foundation

GDPR-Compliant Toolkit

National Reach

THE PRACTICE AT A GLANCE

This national pilot programme launched by the Austrian Ministry of Education in partnership with regional authorities involved one hundred and fourteen schools, selected to reflect balanced representation across regions, school types, and levels of AI experience. The main goal was strategic: moving beyond scattered AI use by systematically assessing teacher AI skills, identifying effective support approaches, and providing a controlled space for educators to experiment with generative AI tools. The programme focused entirely on supporting teachers rather than introducing student-facing tools.

The pilot was part of a broader national strategy: 114 schools were selected from over 350 applications. A scientific evaluation led by the University of Graz used pre- and post-questionnaires based on the "Intelligent TPACK" framework to measure changes in teacher attitudes and AI literacy.

HOW IT IMPROVES TEACHING AND LEARNING

The pilot explored how AI could improve professional practice by supporting lesson planning, content creation, and administrative tasks. The core aim was to assess teachers' attitudes towards AI and build their skills, creating a foundation for future classroom use. Research shows that teacher confidence or anxiety significantly affects how AI is adopted, which justified this teacher-focused approach.

A toolkit developed by the University of Graz research partner prioritised platforms with clear teaching potential and full GDPR compliance. Participating schools chose from tools including Fobizz, an AI-powered platform for educators supporting the creation and delivery of digital learning materials with features like AI chat and image generation; and Teachino, an Austrian-developed lesson planning assistant using AI to help create, organise, and share teaching materials. These tools addressed the challenge of making AI accessible for teachers without requiring technical expertise.

Generative AI for content creation; AI-powered curation and recommendation

KI-Pilotschulen (AI Pilot in Schools)

Austria

Primary and Secondary Education

TEACHER CHALLENGES & INSIGHTS

Administrative Delays

Reaching Beyond Motivated Teachers

Age Restrictions in Primary Schools

“Although part of a well-supported national strategy with schools selected from over three hundred and fifty applications, the project faced significant delays. [...] Despite the constraints, the pilot produced a list of good practices now being prepared for publication to help all schools.”

HOW IT SUPPORTS TEACHERS AND LEARNERS

As part of the broader national AI strategy, a Massive Open Online Course launched in May 2024 to support teacher training across all Austrian schools. The University of Graz helped design the course, which included four compulsory and two optional modules covering AI introduction, ethical issues, and practical classroom support. The course was open to all educators, not just pilot participants.

Within pilot schools, national training was supported by local help from school e-learning coordinators and optional training from tool providers. Many teachers found the tools easy enough to learn through direct use. Participation was voluntary, which ensured an engaged group but may have attracted more motivated educators. In one school interviewed, only ten teachers chose to join, showing the challenge of reaching a wider group.

MAKING IT WORK IN PRACTICE

Although part of a well-supported national strategy with schools selected from over three hundred and fifty applications, the project faced significant delays. Rules preventing credit card payments for licenses compressed the active phase into a very short window between November and December 2024, limiting the ability to gather meaningful data on tool impact. The University of Graz led a scientific evaluation using pre- and post-questionnaires to assess changes in teacher attitudes and AI literacy, within the limits of the short implementation period.

All selected tools were checked for GDPR compliance. Some teachers raised concerns about AI's environmental impact. The pilot faced specific challenges in primary schools due to age restrictions and concerns about introducing AI before foundational computing skills. Despite the constraints, the pilot produced a list of good practices now being prepared for publication to help all schools. Some EU-based tools remained available beyond December in certain schools, with Fobizz accessible for an additional six months and Teachino for the full school year.

Website: https://www.bmb.gv.at/Themen/schule/zrp/ki.html

Dialogue-based tutors; Exploratory learning environments

ROLEPL-AI

Denmark

Adult Education

“Younger students using role-play in English classes responded very positively. [...] Older workers in company classes reacted negatively, largely because of unfamiliar virtual environments rather than AI itself, with many not knowing what virtual reality meant.”

KEY BENEFITS

- **Authentic Practice**
- **Transferable Teacher Skills**
- **Learner-Centred Privacy**

THE PRACTICE AT A GLANCE

ROLEPL-AI is a collaborative project led by French immersive-learning company Manzalab, creating an AI-based role-play simulation in a virtual campus to help learners practise workplace soft skills remotely. The project brings together immersive technology, artificial intelligence, and teaching practice. VUC Storstrøm, a Danish public provider of general adult education that also designs training courses for companies, joined the project to explore how AI-supported simulations can make general education more relevant to employer needs whilst working for ordinary adult groups.

Teachers had to strip content down to machine-readable facts, removing explanations, examples, and scaffolding. As a result, teachers built deep skills in prompting, content creation, and AI literacy now shared with colleagues across the school.

HOW IT IMPROVES TEACHING AND LEARNING

The project responds directly to a clear need: helping adult learners build soft skills for workplace communication. The team mapped the role-play tool onto existing general education courses in communication, collaboration, and digital skills. Teachers worked with developers to create scenarios and characters reflecting local workplaces, making practice feel authentic.

Testing revealed two different learner experiences. Older workers in company classes reacted negatively, largely because of unfamiliar virtual environments rather than AI itself, with many not knowing what virtual reality meant. Younger students using role-play in English classes responded very positively. A clear skills gap emerged: smooth conversation in the tool requires B2-level English, otherwise learners spend their effort on language and basic navigation rather than soft skills practice. Some lower-level learners found workarounds using translation apps, which incidentally built their English and digital skills. This shows the need to assess foundational skills before using such tools.

Dialogue-based tutors; Exploratory learning environments

ROLEPL-AI

Denmark *Adult Education*

TEACHER CHALLENGES & INSIGHTS

Intensive Scenario Design

Language Skills Barrier

High Server Costs

"Even initially resistant groups engaged constructively once positioned as collaborators rather than passive users."

HOW IT SUPPORTS TEACHERS AND LEARNERS

Designing simulations demanded substantial teacher effort unlike preparing usual lessons. Content had to be stripped down to machine-readable facts, removing the explanations, examples, and support teachers normally provide. This hidden preparation cost is specialised, requiring many rounds of revision, often six or seven, with the development team. Using such tools is not a simple time-saver but requires prior knowledge, careful preparation, and sustained effort.

Teachers were active participants, testing tools before classroom use to understand their limits. The school had a trusting relationship with developers and was active in a regional network advising the Ministry of Education. Teacher selection mattered: leads with strong prompting skills were identified, and experimentally minded teachers were chosen for classroom testing. Students were told the tool was under development and their feedback mattered. This co-creator framing changed the dynamic completely. Even initially resistant groups engaged constructively once positioned as collaborators rather than passive users.

MAKING IT WORK IN PRACTICE

Early tests revealed hallucinations, where the AI gave inaccurate or nonsensical responses, which technical partners fixed before wider use. Simple safeguards were added, including a student button to flag if the AI became inappropriate, plus clear boundaries on what it could say. Privacy choices were deliberate: although a teacher dashboard could have allowed monitoring, it was not built due to ethical concerns. Instead, students could choose to download their own conversation records, prioritising learner control over administrative convenience.

The main barrier is cost. Server expenses limit access to short testing windows, preventing regular classroom use and proper impact assessment. Using the tool also requires computers and headsets with microphones. As a product, daily use is not viable under current costs. As a practice, however, the project remains valuable: teachers developed deep skills in prompting, content creation, and AI literacy that are now shared with colleagues, enabling the school to expand its offerings including AI literacy courses even when the original tool is not running.

Website: https://www.rolepl-ai.com/

Dialogue-based tutors; AI-powered curation and recommendation

AI Twins

Lithuania

Higher Education

“Students valued having access to course knowledge at any time, and professors noted that students felt they had more control over their learning.”

KEY BENEFITS

24/7 Expertise Access

Built-In Integrity

Multilingual Support

THE PRACTICE AT A GLANCE

Two lecturers at Vilnius University's Faculty of Law piloted “AI knowledge twins” as course-based teaching assistants for Data Protection Law. Each twin is a chatbot built on the lecturer's own publications and teaching materials, used by students to ask questions, review content, and prepare assignments. The goal was to extend access to instructor expertise around the clock, increase engagement, and support inclusivity in a course designed for international students. Development began when professors realised students were already using general AI tools for assignments, often without guidance.

The final exam requires each student to interrogate the Twin until it hallucinates, then critique the output, turning AI reliability into a learning objective.

HOW IT IMPROVES TEACHING AND LEARNING

The lecturers believe the Twins offer a safer alternative to general AI tools because they draw on teachers' own materials rather than unknown internet sources. Students valued having access to course knowledge at any time, and professors noted that students felt they had more control over their learning. The Twins encourage students to ask questions freely and seek support in a structured way.

Students report that the tool reduces stress and helps them absorb course content better. It saved time and, according to interviews, led to deeper classroom discussions with more thoughtful questions. Students appreciated being able to use AI openly and discuss the results freely. However, they cautioned that over-reliance could weaken independent thinking and limit the variety of ideas when many students ask the same questions.

Because answers come from lecturers' own materials, academic integrity is better protected. Students are encouraged to test the Twins' limits and record any errors during assessment, building critical inquiry into newly developed tools. This links directly to course goals around legal reasoning and responsible technology use.

Dialogue-based tutors; AI-powered curation and recommendation

AI Twins

Lithuania *Higher Education*

TEACHER CHALLENGES & INSIGHTS

Longer Marking Time

Risk of Over-Reliance

Unrecognised Extra Workload

“Students combining work with study confirmed flexibility benefits, with significant time saved when seeking clarification or generating ideas, allowing learning to fit around jobs and family.”

HOW IT SUPPORTS TEACHERS AND LEARNERS

Creating a digital twin is deliberately simple, with uploading content taking around fifteen to thirty minutes. Whilst the Twins help teachers save time by filtering routine questions during term, this is partly offset at assessment time. Marking AI-enabled exams took three or four times longer because answers were longer and included conversation logs.

The Twin works in any language, helping the international student group. Students combining work with study confirmed flexibility benefits, with significant time saved when seeking clarification or generating ideas, allowing learning to fit around jobs and family. Control was deliberately given to learners: lecturers invited classes to choose topics to explore and prepare peer-teaching presentations with AI support. Final exams required each student to push the Twin until it made errors, then critique the output.

MAKING IT WORK IN PRACTICE

The project used a strict approach where students ask questions without logging in and the system stores nothing centrally. Communication stays private and anonymous, making learners noticeably more comfortable than with public chatbots. Students reported feeling more secure about privacy and more confident using a lecturer's Twin than general tools like ChatGPT. Critical thinking about AI is built into the teaching itself, with exams requiring students to challenge tools to the point of error, turning reliability questions into learning goals.

The platform accepts documents in any language, serving multilingual groups without extra technical work. To spread the approach, the team produced short guidelines and offered coaching calls for interested colleagues. The professor expects rapid growth, predicting that every academic will eventually have their own digital twin. Costs are low, mainly staff time. Remaining challenges include clearer privacy communication for students, automatic source lists to help fact-checking, and formal recognition of the extra workload early adopters take on.

Website: https://www.tf.vu.lt/news

Dialogue-based tutors; Exploratory learning environments

Talk2Transform

Germany *Higher Education*

“This safe space boosted engagement, with faculty surprised by how thoroughly students prepared.”

KEY BENEFITS

Immediate Feedback

Risk-Free Practice

Scalable Learning

THE PRACTICE AT A GLANCE

Talk2Transform is an AI-supported conversation simulation tool developed at Brandenburg University of Technology for business management students. The tool is a custom AI assistant based on a model like GPT-4o, set up with specific instructions. Its main purpose is helping students practise leadership communication by interacting with virtual employees to solve problems, give feedback, set goals, and motivate team members. The project grew from a faculty member's personal curiosity about AI's potential to extend teaching capacity beyond what traditional classrooms allow.

Development from idea to classroom use took approximately 2.5 months with a team of five to six people, none with IT backgrounds. The speed was attributed to bypassing typical administrative and funding processes.

HOW IT IMPROVES TEACHING AND LEARNING

Talk2Transform addresses a real gap in business education: students need strong communication and leadership skills, but traditional teaching offers limited chances to practise. This tool allows individualised practice with large numbers of students, something impossible in conventional role-play with thirty people over several rounds.

Students found the chance to simulate realistic conversations and receive immediate, system-generated feedback particularly helpful, describing it as good preparation for real-world situations and future careers. They reported high satisfaction with the learning effect, finding the tool engaging and relevant. Many said work with the chatbot was useful and important, noting that their communication skills improved. A key benefit was practising without fear of judgement, with time to think and take notes unlike face-to-face interactions. This safe space boosted engagement, with faculty surprised by how thoroughly students prepared. Beyond skill building, Talk2Transform aims to develop critical understanding of AI itself, helping students grasp how it works, its limits, and appropriate use.

Talk2Transform

Germany

Higher Education

TEACHER CHALLENGES & INSIGHTS

- **Initial Setup and Scenario Creation**
- **Technical Issues (Crashes, Slow Performance)**
- **Maintaining Trust with Students**

"Student feedback was overwhelmingly positive, emphasising usefulness for future careers. [...] Some felt nervous in face-to-face conversations but found AI helped prepare them by creating challenging scenarios to work through."

HOW IT SUPPORTS TEACHERS AND LEARNERS

Whilst the tool offers practice opportunities for large groups, initial setup and scenario creation required significant time and planning. Faculty noted that lecturers often lack allocated time to develop innovative teaching formats, making it partly a personal project rather than fully recognised work. This echoes findings from other universities about systematic challenges in supporting teaching innovation.

The team emphasised maintaining trust with students by acknowledging this was an experiment and assuring them that technical problems would not affect grades. This proved important as Talk2Transform experienced crashes, slow performance, and inconsistent responses. Students also reported technical issues including delays and page errors. Trust contributed to acceptance, with no resistance, especially since many students already knew AI tools.

Student feedback was overwhelmingly positive, emphasising usefulness for future careers. Students described AI conversations as valuable experiences and interesting ways to learn. Some felt nervous in face-to-face conversations but found AI helped prepare them by creating challenging scenarios to work through.

MAKING IT WORK IN PRACTICE

Talk2Transform developed organically through personal interest and commitment of a small team of around five to six people with varied expertise including content writing and emotional design, though none had IT backgrounds. Development from idea to use took about two and a half months, with speed attributed to bypassing typical administrative and funding processes. Support from university leadership, especially the Vice-President for Teaching, proved crucial. The small university size allowed easy communication and team formation.

The text-based chat approach rather than complex avatar systems makes scaling within the institution possible. The tool gained national recognition as a best practice example, with findings shared at conferences and in publications. Plans include refining feedback and potentially integrating the simulator into larger cybersecurity training research. Sustainability depends heavily on dedicated time for lecturers. The team stressed that universities must provide institutional support for innovation rather than relying on personal motivation and work outside normal hours.